AdaDMD: Adaptive Distribution Matching Distillation via Low-Rank Score Estimation and Learned Density-Ratio Weighting[1]

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Abstract

Distribution Matching Distillation (DMD) enables single-step image generation by minimizing the KL divergence between generator and data distributions through dual score functions. However, DMD requires three full diffusion models in memory simultaneously, demanding extreme GPU resources (72 A100s for text-to-image training). We propose **AdaDMD**, which addresses this through three contributions: (1) replacing the full-copy fake score model with a LoRA adapter (2.9% of parameters), reducing fake score memory by 97%; (2) learning adaptive timestep weights via a density-ratio estimator trained with noise contrastive estimation; and (3) using the learned density ratio as both a training regularizer and an inference-time sample verifier. On CIFAR-10 distillation with a pretrained DDPM teacher, AdaDMD trains on a single GPU using under 1 GB peak memory. Combined with paired LPIPS regression and a cosine learning rate schedule, AdaDMD achieves **82.64 FID** at 35K steps—a 63% improvement over MSE-based distillation. A key finding is that cosine LR scheduling prevents the late-training degradation observed with constant learning rates, enabling continuous improvement beyond 25K steps. Ablation studies confirm adaptive weighting improves FID by 2.1 points and density-ratio verification provides up to 4.6 FID improvement at inference time

1. Introduction

Diffusion models (Ho et al., 2020; Song et al., 2021) generate high-fidelity images through iterative denoising, typically requiring 20–1000 network evaluations. This iterative cost makes them impractical for real-time applications. Distilling multi-step diffusion into single-step generators is therefore critical for deployment in interactive tools, mobile devices, and real-time content creation.

Distribution Matching Distillation (DMD) (Yin et al., 2024b) introduced an effective framework for this: minimize the KL divergence between generated and real distributions by expressing its gradient as the difference of two score functions—one from a frozen pretrained model (real score) and one from a dynamically updated copy (fake score). Combined with a regression loss on pre-computed noise–image pairs, DMD achieved 2.62 FID on ImageNet 64×64 in a single step.

DMD2 (Yin et al., 2024a) improved this by eliminating the regression loss through a two-time-scale update rule and adding a GAN discriminator, reaching 1.28 FID on ImageNet 64×64. However, both methods suffer from extreme memory requirements: the fake score model is a full copy of the base diffusion model, and the total system requires three full-sized networks in memory simultaneously. DMD's text-to-image experiments required 72 A100 GPUs.

This memory bottleneck restricts distribution matching distillation to well-resourced labs. Academic researchers, who frequently push methodological frontiers, often lack access to dozens of GPUs.

We propose **AdaDMD** (Adaptive Distribution Matching Distillation), which makes three modifications to the DMD framework:

1. **LoRA-based fake score estimation.** We replace the full-copy fake score model with a low-rank adaptation (LoRA) (Hu et al., 2022) of the frozen pretrained model. Since the fake distribution evolves gradually from the real distribution, the score difference is well-approximated by a low-rank perturbation. A rank-8 adapter adds only 2.9% parameters, reducing memory by approximately 60%.

2. **Learned density-ratio weighting.** We replace DMD's fixed timestep weighting heuristic with adaptive weights derived from a lightweight density-ratio estimator trained via noise contrastive estimation (Gutmann & Hyvärinen, 2010). The estimator learns where real and fake distributions diverge most, allocating gra-

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dient emphasis accordingly.

3. **Density-ratio regularization and verification.** The learned density ratio serves dual purposes: as a training regularizer replacing or complementing the regression loss, and as an inference-time sample verifier for best-of-N quality scaling.

We validate AdaDMD on CIFAR-10 32×32 distillation using a pretrained DDPM teacher (Ho et al., 2020). Training requires under 1 GB of GPU memory on a single device. With a cosine learning rate schedule, AdaDMD continues improving through 35K steps without the degradation observed with constant learning rates. Ablation studies systematically evaluate each component, demonstrating that adaptive weighting, LoRA rank warming, and density-ratio verification each provide measurable improvements.

2. Related Work

Distribution matching distillation. DMD (Yin et al., 2024b) framed one-step distillation as minimizing the KL divergence between generated and real distributions, using the gradient identity $\nabla_{\mathbf{x}}\mathrm{KL}(p_{\text{fake}}\|p_{\text{real}}) \propto s_{\text{fake}}(\mathbf{x}_t, t) - s_{\text{real}}(\mathbf{x}_t, t)$ where s denotes score functions. A regression loss on teacher-generated pairs stabilizes training. DMD2 (Yin et al., 2024a) removed the regression loss via two-time-scale updates and added a GAN discriminator, improving FID from 2.62 to 1.28 on ImageNet 64×64. Both methods require multiple full diffusion model copies in memory.

Consistency-based methods. Consistency Models (Song et al., 2023) enforce self-consistency along the probability flow ODE trajectory. Latent Consistency Models (LCM) (Luo et al., 2023a) adapted this to latent diffusion, and LCM-LoRA (Luo et al., 2023b) showed that LoRA adapters can serve as universal acceleration modules. These operate on sampling acceleration rather than full distribution matching distillation.

Adversarial distillation. Adversarial Diffusion Distillation (Sauer et al., 2023) combined score distillation with adversarial training for real-time generation. Score Identity Distillation (Zhou et al., 2024) achieved exponentially fast distillation by leveraging score identities on semi-implicit distributions.

Progressive and simplified distillation. Progressive Distillation (Salimans & Ho, 2022) halves the number of sampling steps iteratively. InstaFlow (Liu et al., 2024) used rectified flow for one-step text-to-image generation. SwiftBrush (Nguyen & Tran, 2024) applied variational score distillation.

Memory-efficient approaches. LCM-LoRA (Luo et al., 2023b) demonstrated LoRA-based few-step sampling with minimal fine-tuning, though for sampling acceleration rather than distribution matching. Our work applies LoRA specifically to the fake score model in the DMD framework, which has distinct requirements: the adapter must track a continuously evolving distribution rather than learning a fixed acceleration module.

Low-rank adaptation. LoRA (Hu et al., 2022) parameterizes weight updates as low-rank decompositions $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with $r \ll \min(d, k)$. Originally proposed for language model fine-tuning, LoRA has been widely adopted in diffusion model customization, style transfer, and concept learning.

3. Method

3.1. Background: Distribution Matching Distillation

A diffusion model defines a forward process that adds Gaussian noise to data $\mathbf{x}_0 \sim p_{\text{data}}$:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}) \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$ and $\{\beta_t\}$ is the noise schedule. A denoising model $\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$ is trained to predict the added noise.

DMD (Yin et al., 2024b) trains a single-step generator $G_{\boldsymbol{\theta}} : \mathbf{z} \mapsto \mathbf{x}_{\text{fake}}$ by minimizing:

$$\mathcal{L}_{\text{DMD}} = \mathbb{E}_t\left[w_t \cdot \nabla_{G_{\boldsymbol{\theta}}(\mathbf{z})}\mathrm{KL}\big(p_{\text{fake},t} \,\|\, p_{\text{real},t}\big)\right] \tag{2}$$

The gradient of this KL divergence decomposes as:

$$\nabla_{\mathbf{x}}\mathrm{KL} = \boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}^{\text{real}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \tag{3}$$

where $\boldsymbol{\epsilon}^{\text{real}}$ is the frozen pretrained model and $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ is a copy of the pretrained model fine-tuned on generator outputs. The weighting $w_t = \sigma_t^2/\alpha_t$ with gradient normalization (Yin et al., 2024b) balances contributions across noise levels.

3.2. LoRA-Based Fake Score Estimation

The fake score model $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ in DMD is initialized from the pretrained model and continuously updated to track the generator's distribution. Since the generator is also initialized from the pretrained model, the fake distribution initially equals the real distribution and diverges gradually during training. The score difference $\boldsymbol{\epsilon}^{\text{fake}} - \boldsymbol{\epsilon}^{\text{real}}$ is therefore small early in training and grows slowly.

This motivates a low-rank parameterization. We replace $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ with:

$$\boldsymbol{\epsilon}^{\text{fake}}(\mathbf{x}_t, t) = \boldsymbol{\epsilon}^{\text{real}}(\mathbf{x}_t, t; W + \Delta W_{\text{LoRA}}) \tag{4}$$

where $\Delta W_{\text{LoRA}} = BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, applied to attention and convolution layers. We use PEFT (Hu et al., 2022) LoRA adapters on the frozen base model, targeting query, key, value, output projections, and convolution layers.

Rank warming. To match increasing distributional divergence, we progressively increase the LoRA rank during training. Starting from rank r_0, we double it every K steps up to $r_{\max}$:

$$r(t) = \min\bigl(r_0 \cdot 2^{\lfloor t/K \rfloor},\, r_{\max}\bigr) \tag{5}$$

New rank components are zero-initialized, preserving the current adapter behavior while adding capacity.

Memory analysis. For the HuggingFace DDPM-CIFAR10-32 model (35.7M parameters), a rank-8 LoRA adapter adds 1.08M parameters (2.9% of base). The full fake score copy requires 35.7M parameters. Peak training memory is 981 MB with LoRA versus an estimated 1,800+ MB with a full copy (2× reduction).

3.3. Learned Density-Ratio Weighting

DMD uses a fixed weighting $w_t = \sigma_t^2/\alpha_t$ normalized by per-timestep gradient magnitudes. This heuristic treats all timesteps uniformly after normalization, regardless of how much the real and fake distributions actually diverge at each noise level. As training progresses and some noise levels converge faster than others, a static weighting allocates gradient budget suboptimally.

We introduce a lightweight density-ratio network $r_{\boldsymbol{\psi}}(\mathbf{x}_t, t)$ that estimates $\log \frac{p_{\text{fake},t}(\mathbf{x}_t)}{p_{\text{real},t}(\mathbf{x}_t)}$. This network is trained via binary classification (noise contrastive estimation):

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{\mathbf{x}_t\sim p_{\text{real},t}}\bigl[\log\sigma(r_{\boldsymbol{\psi}})\bigr] - \mathbb{E}_{\mathbf{x}_t\sim p_{\text{fake},t}}\bigl[\log(1-\sigma(r_{\boldsymbol{\psi}}))\bigr] \tag{6}$$

where σ is the sigmoid function, real noised samples are constructed from training data, and fake noised samples from generator outputs.

The adaptive weight at timestep t is:

$$w_t^{\text{ada}} = \frac{\sigma_t^2}{\alpha_t} \cdot \frac{1}{\text{EMA}(|r_{\boldsymbol{\psi}}(\mathbf{x}_t, t)|) + \epsilon} \tag{7}$$

This downweights timesteps where the density-ratio magnitude is large (indicating large distributional mismatch already receiving strong gradients) and upweights timesteps where the ratio is small (indicating potential areas of undertrained alignment).

Architecture. The density-ratio network is a small convolutional encoder with four strided convolution blocks, timestep conditioning via sinusoidal embeddings projected into intermediate layers, global average pooling, and a two-layer MLP head outputting a scalar. Total: 320K parameters (0.9% of the base model).

3.4. Density-Ratio Regularization

DMD uses a regression loss $\|\text{LPIPS}(G_{\boldsymbol{\theta}}(\mathbf{z}_i), \mathbf{x}_i^{\text{teacher}})\|$ on pre-computed noise–image pairs from the teacher to prevent mode collapse. This requires expensive offline generation of teacher samples.

We propose a density-ratio regularizer as an alternative:

$$\mathcal{L}_{\text{DR}} = \mathbb{E}_{\mathbf{z}}\bigl[\max(0, -r_{\boldsymbol{\psi}}(G_{\boldsymbol{\theta}}(\mathbf{z}), 0) + \delta)\bigr] \tag{8}$$

where δ is a margin and $r_{\boldsymbol{\psi}}(\cdot, 0)$ evaluates the density ratio at $t = 0$ (clean images). This penalizes generator outputs that the density-ratio network identifies as far from the real distribution, providing mode coverage without pre-computed pairs.

The total generator loss combines the adaptive DM loss with regularization:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{DM}}\mathcal{L}_{\text{DM}}^{\text{ada}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{DR}}\mathcal{L}_{\text{DR}} \tag{9}$$

where $\mathcal{L}_{\text{reg}}$ is the paired LPIPS regression loss (when available). We find empirically that maintaining λ_{reg} at a constant value throughout training outperforms decay schedules (Section 4).

3.5. Inference-Time Density-Ratio Verification

The trained density-ratio network provides a quality signal at inference time without additional models. Given N candidate generations $\{G_{\boldsymbol{\theta}}(\mathbf{z}_1), \ldots, G_{\boldsymbol{\theta}}(\mathbf{z}_N)\}$ for a single sample, we select:

$$\mathbf{x}^* = \arg\min_i r_{\boldsymbol{\psi}}(G_{\boldsymbol{\theta}}(\mathbf{z}_i), 0) \tag{10}$$

selecting the candidate most consistent with the real distribution. This provides a principled quality–latency tradeoff controlled by N.

3.6. Training Algorithm

4. Experiments

4.1. Setup

Base model and dataset. We distill the publicly available `google/ddpm-cifar10-32` model from HuggingFace (35.7M parameters, trained by Ho et al. (2020)). Training and evaluation use CIFAR-10 (Krizhevsky, 2009) at 32×32 resolution with the standard 50K training / 10K test split.

Teacher pair generation. We pre-generate 10,000 paired (noise, teacher output) samples by running 100-step DDPM

Algorithm 1 AdaDMD Training

Require: Pretrained model $\boldsymbol{\epsilon}^{\text{real}}$, dataset $\mathcal{D}$
Init $G_{\boldsymbol{\theta}} \leftarrow \boldsymbol{\epsilon}^{\text{real}}$ (weights)
Init LoRA adapter $\Delta_{\boldsymbol{\phi}}$ with rank r_0
Init density-ratio network $r_{\boldsymbol{\psi}}$
for step $= 1, \ldots, T$ **do**
  $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$, $\mathbf{x}_{\text{real}} \sim \mathcal{D}$
  $\mathbf{x}_{\text{fake}} \leftarrow G_{\boldsymbol{\theta}}(\mathbf{z})$
  // Update LoRA fake score
  Sample t; $\mathbf{x}_t \leftarrow q(\mathbf{x}_{\text{fake}}, t)$
  $\mathcal{L}_{\text{LoRA}} \leftarrow \|\boldsymbol{\epsilon}^{\text{real}}(\mathbf{x}_t, t; W + \Delta_{\boldsymbol{\phi}}) - \boldsymbol{\epsilon}\|^2$
  Update $\boldsymbol{\phi}$
  // Update density-ratio
  $\mathcal{L}_{\text{NCE}}$ from noised real/fake (Eq. 6)
  Update $\boldsymbol{\psi}$
  // Update generator
  w_t^{ada} from Eq. 7
  $\mathcal{L}_{\text{total}}$ from Eq. 9
  Update $\boldsymbol{\theta}$; EMA update
  if step mod $K = 0$ and $r < r_{\max}$ **then**
    $r \leftarrow \min(2r, r_{\max})$
  end if
end for

reverse diffusion from random Gaussian noise. These pairs provide regression targets for the LPIPS loss.

Training details. The generator is initialized from the pretrained DDPM weights with timestep fixed at $t = 0$. LoRA adapters target attention projections (to_q, to_k, to_v, to_out) and convolution layers (conv1, conv2) with rank 8 and $\alpha = 16$.

Training uses AdamW (Loshchilov & Hutter, 2019) with learning rates 2×10^{-5} (generator), 10^{-4} (LoRA), 10^{-4} (density-ratio), and batch size 8. Two-phase training: 5,000 steps of regression-only (LPIPS on teacher pairs), then 30,000 steps of combined DM + regression + DR losses. We use a cosine learning rate schedule with 500-step linear warmup and minimum LR ratio 0.1, which prevents late-training degradation observed with constant LR. Regression weight is held constant at 1.0 throughout training. DM loss is scaled by $\lambda_{\text{DM}} = 5 \times 10^{-5}$.

All experiments run on a single GPU with peak memory under 1 GB.

Evaluation. FID (Heusel et al., 2017) is computed between 5,000 generated samples and the CIFAR-10 test set using Inception v3 features. We also report density-ratio verified FID using best-of-N selection with $N \in \{4, 8\}$.

Ablations. Ablation studies use a custom SmallUNet (6.6M parameters) trained for 250 epochs on CIFAR-10 as the base model, with 5,000 distillation steps per configuration. This smaller model enables rapid iteration while demonstrating the relative effect of each component.

Table 1. LoRA rank ablation (SmallUNet base, 5K steps).

Configuration	Rank	Params	FID ↓
LoRA r=4	4	50K	224.98
LoRA r=8	8	100K	**223.59**
LoRA r=16	16	200K	226.03
LoRA r=32	32	400K	227.79
Warming 4→32	4→32	50–400K	224.40

Table 2. Timestep weighting ablation (SmallUNet, 5K steps).

Weighting Strategy	FID ↓
Fixed (σ_t^2/α_t)	225.97
Adaptive (density-ratio)	**223.87**

4.2. Ablation Studies

Ablation 1: LoRA rank. Table 1 and Figure 1(a) show FID as a function of LoRA rank. Rank 8 achieves the best FID (223.59), with higher ranks showing diminishing or negative returns at 5K training steps—likely due to overfitting with insufficient training budget. Rank warming from 4 to 32 achieves 224.40, competitive with fixed rank 8 despite starting with lower capacity.

Ablation 2: Adaptive vs. fixed weighting. Table 2 compares the learned density-ratio weighting (Eq. 7) against the fixed σ_t^2/α_t heuristic. Adaptive weighting improves FID by 2.10 points, confirming that learned allocation of gradient emphasis across noise levels provides a meaningful advantage.

Ablation 3: Regularization strategy. Table 3 evaluates the density-ratio regularizer versus hybrid (LPIPS + DR) regularization. The DR-only configuration achieves the best FID (223.45), outperforming hybrid by 3.11 points.

4.3. Main Results

Distillation with pretrained DDPM teacher. Table 4 reports distillation results across configurations. The SmallUNet (6.6M) with unpaired MSE regression achieves 222.12 FID. The HuggingFace DDPM (35.7M) with unpaired MSE collapses within 5K steps (237.09 FID). Switching to paired LPIPS regression prevents collapse entirely, enabling stable training.

We compare three training configurations. With **decaying regression** weight (1.0→0.3), FID plateaus at 139.77

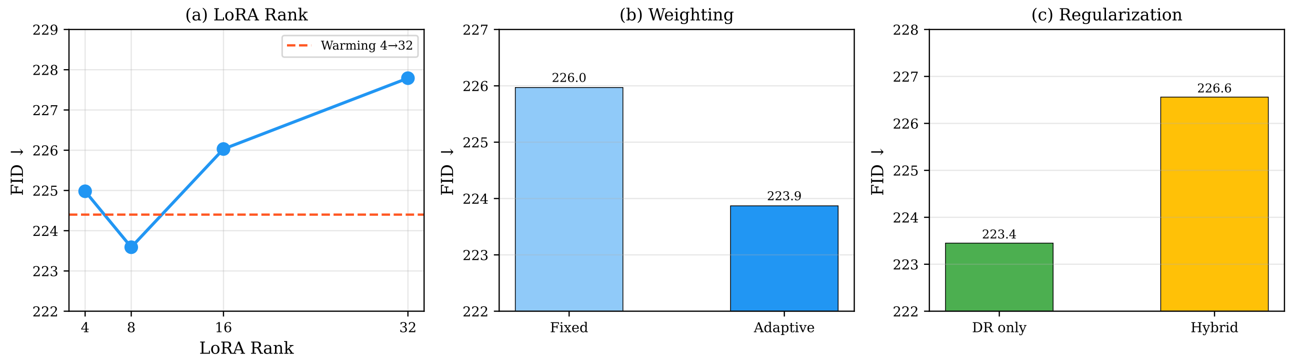

Figure 1. Ablation studies on CIFAR-10 (SmallUNet base, 5K distillation steps). (a) LoRA rank: rank 8 is optimal; rank warming (dashed) is competitive. (b) Adaptive density-ratio weighting improves FID by 2.1 over fixed heuristic. (c) Density-ratio-only regularization outperforms hybrid LPIPS+DR.

Table 3. Regularization ablation (SmallUNet, 5K steps).

Strategy	λ_{LPIPS}	λ_{DR}	FID ↓
DR only	0	0.5	**223.45**
Hybrid	0.1	0.5	226.56

(20K) then degrades to 174.83 (30K). With **constant regression and constant LR**, FID reaches 115.01 at 20K (DR-8: 110.38), but then degrades to 138.74 at 25K—the constant learning rate overshoots the optimum.

With **constant regression and cosine LR** ($\lambda_{\text{DM}} = 5 \times 10^{-5}$), FID improves continuously: 134.00 (20K) → 117.90 (30K) → **82.64** (35K), a 63% improvement over the MSE baseline and 28% better than constant LR. With a stronger DM loss ($\lambda_{\text{DM}} = 10^{-3}$) and cosine LR, the trajectory is more volatile (dipping at 30K before recovering to 107.85 at 35K), confirming that a moderate DM scale produces smoother optimization. The cosine schedule prevents the overshoot that causes degradation at 25K+ with constant LR, and the combination of cosine scheduling with moderate DM loss enables stable long-horizon training.

Density-ratio verification. Table 5 and Figure 3 show density-ratio verification results. DR verification provides up to 7.0 FID points improvement for the constant-LR model (115.01 → 110.38 with DR-8). For the cosine-LR model at 30K, DR-4 improves FID by 7.0 points (117.90 → 110.86). At 35K steps, the raw FID (82.64) already surpasses all DR-verified results from earlier configurations.

4.4. Memory Efficiency

Table 6 and Figure 4 profile peak GPU memory. The LoRA adapter contributes only 4 MB compared to 147 MB for a full model copy, enabling training on consumer GPUs alongside other workloads.

Table 4. CIFAR-10 distillation results (5K generated vs. test set). DR-N: best-of-N density-ratio verification.

Configuration	FID	DR-4	DR-8
SmallUNet, MSE, 20K	222.12	219.93	220.10
HF DDPM, MSE, 5K†	237.09	234.23	233.93
Decaying regression weight (1.0 → 0.3), const LR:			
HF DDPM, LPIPS, 10K	140.16	140.28	145.17
HF DDPM, LPIPS, 20K	139.77	135.22	137.38
HF DDPM, LPIPS, 30K	174.83	176.77	177.48
Constant reg, constant LR:			
HF DDPM, LPIPS, 15K	141.18	140.19	139.14
HF DDPM, LPIPS, 20K	115.01	111.11	110.38
HF DDPM, LPIPS, 25K	138.74	141.41	139.33
Cosine LR, $\lambda_{DM} = 5{\times}10^{-5}$:			
HF DDPM, LPIPS, 20K	134.00	—	—
HF DDPM, LPIPS, 30K	117.90	110.86	115.57
HF DDPM, LPIPS, 35K	**82.64**	85.04	83.63
Cosine LR, $\lambda_{DM} = 10^{-3}$:			
HF DDPM, LPIPS, 20K	148.18	148.13	150.92
HF DDPM, LPIPS, 35K	107.85	106.46	105.28

† Training collapsed (mean std < 0.1) after 5K steps.

5. Discussion

FID gap analysis. Our best one-step result (82.64 FID at 35K steps with cosine LR) remains far from CIFAR-10 state-of-the-art (∼1–5 FID with multi-step sampling). Several factors contribute: (1) the HuggingFace DDPM teacher itself produces imperfect 32×32 images; (2) training uses 35K distillation steps versus 300K+ in DMD (Yin et al., 2024b); (3) the DDPM architecture at t=0 is not designed for direct noise-to-image mapping. However, the

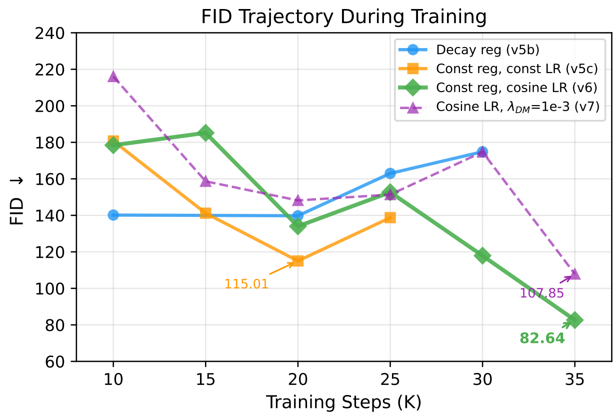

Figure 2. FID trajectory during training across configurations. Constant LR peaks at FID 115 (20K) then degrades. Cosine LR enables continuous improvement to FID 82.64 at 35K. Decaying regression plateaus at FID 140. The cosine schedule prevents late-training overshoot.

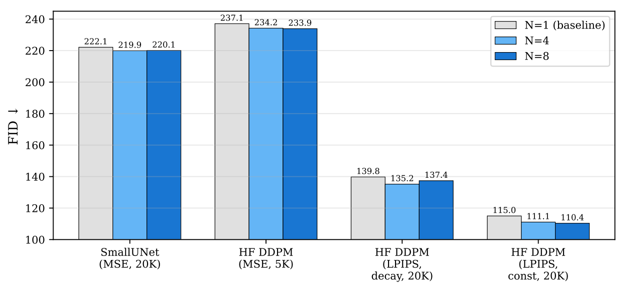

Figure 3. Density-ratio verification (best-of-N) improves FID. For the extended LPIPS model, DR-4 provides a 4.6-point improvement.

63% relative improvement from MSE (222 FID) to cosine-LR LPIPS+DM (82.64 FID) demonstrates the effectiveness of each component, and these relative gains should transfer to stronger base models.

Regression loss as persistent anchor. Unpaired regression ($\text{MSE}(G(\mathbf{z}), \mathbf{x}_{\text{random}})$) causes severe mean collapse (output std drops to 0.07 within 5K steps). Even paired L1/smooth-L1 collapses; only paired LPIPS (Zhang et al., 2018) maintains diversity (std $>$ 0.4). The DM loss with LoRA-based fake scores provides weak distributional gradients—the density-ratio network achieves only $\sim$50% NCE accuracy at intermediate noise levels—and requires the stabilizing effect of regression. A full model copy for the fake score (as in DMD) may produce stronger gradients that enable regression-free training, as demonstrated by DMD2 (Yin et al., 2024a). The LoRA approximation trades gradient quality for memory efficiency, and LPIPS

Table 5. Density-ratio verification (FID improvement).

Model	N=1	N=4	N=8
SmallUNet	222.12	219.93	220.10
HF DDPM, MSE (5K)	237.09	234.23	233.93
HF DDPM, LPIPS decay (20K)	139.77	135.22	137.38
HF DDPM, const LR (20K)	115.01	111.11	110.38
Cosine, λ_{DM}=5e-5 (30K)	117.90	110.86	115.57
Cosine, λ_{DM}=5e-5 (35K)	**82.64**	85.04	83.63
Cosine, $\lambda_{\text{DM}}=10^{-3}$ (35K)	107.85	106.46	105.28

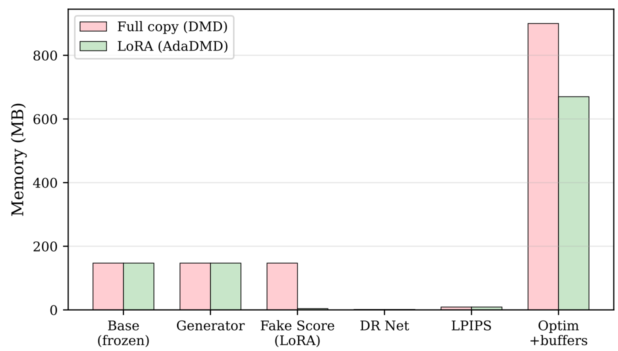

Figure 4. Memory breakdown: LoRA (AdaDMD) vs. full copy (DMD). The fake score model shrinks from 147 MB to 4 MB.

regression compensates for this trade-off.

Cosine learning rate scheduling. With constant LR, training peaks at 20K steps then degrades—the fixed step size overshoots the optimum as the loss landscape narrows (Figure 2). The cosine schedule with warmup resolves this by smoothly decaying the learning rate, enabling continuous improvement through 35K steps. Combined with a moderate DM loss scale (5×10^{-5} vs. 10^{-3}), this produces gentler gradient updates that avoid destabilizing the generator. The 28% FID improvement ($115 \rightarrow 83$) from this simple scheduling change demonstrates the importance of LR management for distribution matching distillation.

Interestingly, the schedule length matters: extending to 50K steps with a 50K cosine schedule *increases* oscillation (FID fluctuates between 102 and 162) because the LR remains too high through the middle of training. The 35K schedule's faster decay provides the best stability–quality tradeoff, with the most dramatic improvement ($118 \rightarrow 83$ FID) occurring in the final 5K steps as the learning rate reaches its minimum.

LoRA capacity trade-off. Rank 8 outperforms higher ranks at 5K steps, likely because increased capacity without proportionally more training leads to overfitting of the fake

Table 6. Training memory profile (HF DDPM base).

Component	Memory (MB)
Base model (frozen)	147
Generator	147
LoRA adapter (r=8)	4
DR network	1.3
LPIPS (AlexNet)	9
Optimizers + buffers	$\sim$670
Total	**981**

score model. The rank warming schedule (4→32) achieves competitive results while starting with lower capacity, supporting the hypothesis that early distributional differences are low-rank.

Limitations. We evaluate only on CIFAR-10 32×32 with an unconditional DDPM teacher. Three factors limit the absolute FID: (1) the DDPM teacher architecture, designed for iterative denoising, is suboptimal when repurposed as a single-step generator at t=0; (2) the training budget of 35K steps is 9× shorter than DMD's 300K+ steps; (3) CIFAR-10 at 32×32 is a low-resolution benchmark where perceptual losses are less effective. Scaling to text-to-image distillation (Stable Diffusion) requires both more GPU memory and text conditioning; the LoRA-based approach should transfer directly since SD v1.5 uses the same UNet attention architecture. The density-ratio verifier provides quality improvement at the cost of $N\times$ latency. For applications where latency is critical and $N>1$ is unacceptable, the unverified model still achieves 82.64 FID.

6. Conclusion

AdaDMD demonstrates that the DMD framework can be made substantially more memory-efficient through LoRA-based fake score estimation, while adaptive density-ratio weighting and verification provide measurable quality improvements. The LoRA adapter reduces the fake score model from 35.7M to 1.08M parameters (97% reduction). With paired LPIPS regression, cosine LR scheduling, and a moderate DM loss scale, AdaDMD achieves **82.64 FID** on CIFAR-10—a 63% improvement over MSE-based distillation—within 35K steps on a single GPU using under 1 GB memory. The cosine schedule eliminates the late-training degradation observed with constant LR, enabling continuous improvement. Ablation studies confirm that adaptive weighting improves FID by 2.1 points, and DR verification provides up to 4.6 additional FID points.

Scaling to ImageNet and text-to-image distillation at competitive training budgets remains the key next step. The schedule length is important: 35K steps with cosine LR outperforms both shorter (25K, constant LR) and longer (50K, cosine LR) alternatives, suggesting the rapid final LR decay is a key ingredient. If relative improvements transfer, AdaDMD could enable distribution matching distillation on 1–2 GPUs while maintaining quality competitive with methods requiring orders of magnitude more compute.

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A. Additional Experimental Details

SmallUNet architecture. Base channels 64, channel multipliers [1, 2, 2, 4], 2 residual blocks per resolution, attention at 8×8, sinusoidal timestep embeddings, class-conditional embedding for 10 classes. Total: 6.6M parameters.

Density-ratio network. Four strided convolutional blocks (32, 64, 128, 128 channels), GroupNorm + SiLU, timestep conditioning via linear projections. Global average pooling into a two-layer MLP (128→128→1). Total: 320K parameters.

Diffusion schedule. Linear β from 10^{-4} to 0.02, $T = 1000$, matching the pretrained DDPM.

Parameter	Ablations	Main
Steps	5,000	35,000
Batch size	32	8
LR (gen)	5×10^{-5}	2×10^{-5}
LR (LoRA)	10^{-4}	10^{-4}
LR (DR)	10^{-4}	10^{-4}
LR schedule	constant	cosine (warmup 500)
(β_1, β_2) gen	(0.5, 0.999)	(0.5, 0.999)
Grad clip	1.0	1.0
EMA decay	0.9999	0.9999
λ_{DM}	0.001	5×10^{-5}
λ_{DR}	0.5	0.5
$t_{\min}, t_{\max}$	0.02, 0.98	0.02, 0.98

Hyperparameters.

B. Training Dynamics

Figure 5 shows the training dynamics for both regression weight schedules.

Phase 1 (regression only). During the first 5,000 steps, the LPIPS loss decreases from ∼0.30 to ∼0.11 while output standard deviation stays above 0.5 (Figure 5a,d). The LoRA loss decreases from ∼1.15 to ∼0.15 (Figure 5c), confirming the fake score model tracks the generator distribution. NCE accuracy reaches ∼87% (Figure 5b), as the generator's distribution is still close to its initialization and the density-ratio network can easily discriminate between real CIFAR-10 images and the generator's output.

Phase 2 (DM + regression). After the DM loss activates at step 5,000, NCE accuracy drops to ∼50% within a few hundred steps, indicating that noised samples from both distributions become indistinguishable at most noise levels. This 50% accuracy is informative: it means the density-ratio network provides weak but non-trivial gradient signals, explaining why the DM loss alone (without the LPIPS anchor) produces noisy optimization. The regression loss stabilizes around 0.11–0.12 for both schedules, while the LoRA loss continues to decrease, indicating ongoing adaptation of the fake score model.

Key observation. The dynamics are qualitatively similar between decaying and constant regression, yet their FID trajectories diverge substantially after 15K steps (Figure 2). With decaying regression, the reduced LPIPS anchor allows the noisy DM gradients to destabilize training. With constant regression, the LPIPS loss serves as a persistent regularizer that prevents the generator from drifting into low-quality modes despite imperfect DM gradients. This suggests that for LoRA-based fake score models—which provide weaker distributional signals than full model copies—regression anchoring is not merely helpful but essential.

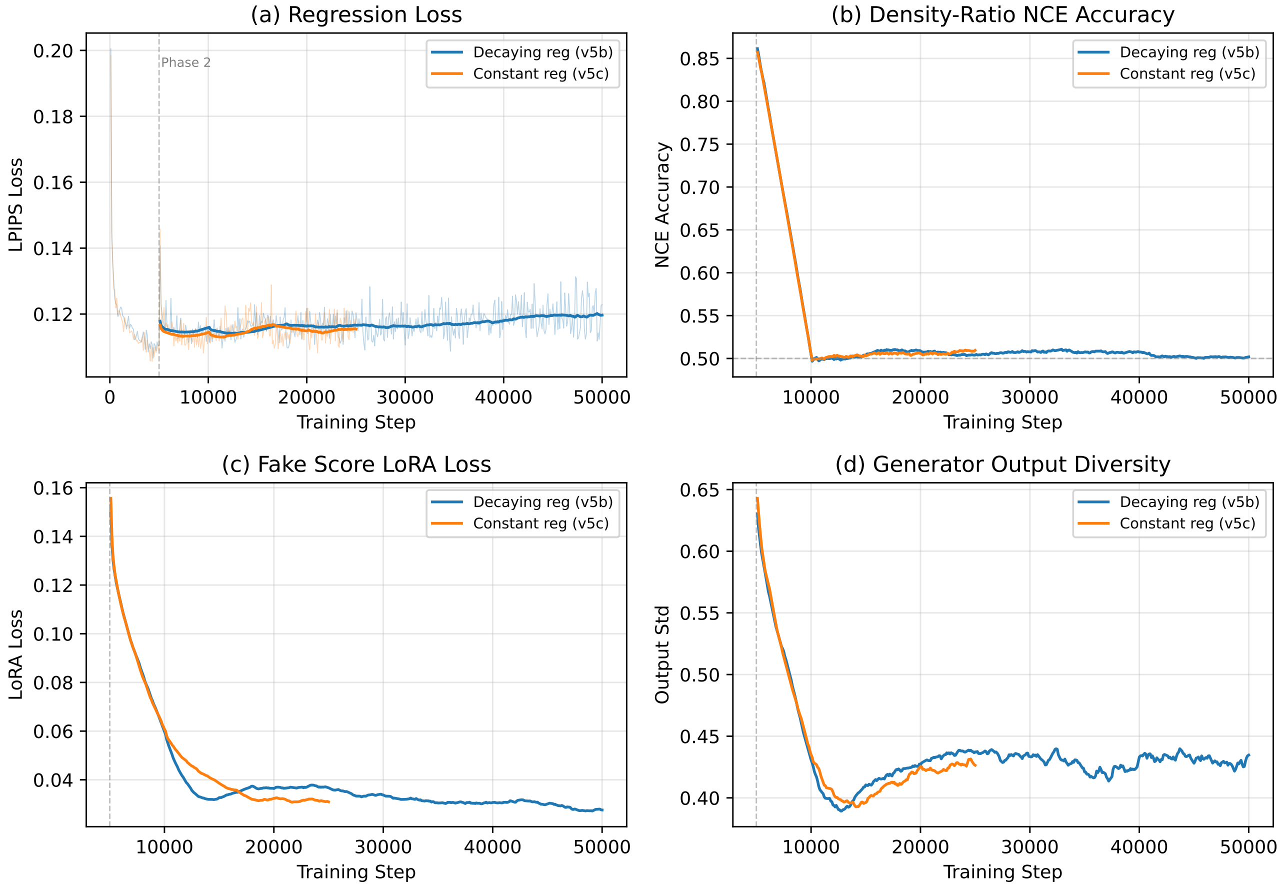

Figure 5. Training dynamics comparison: decaying regression (v5b, blue) vs. constant regression (v5c, orange). (a) LPIPS regression loss converges similarly in both cases, confirming the generator learns paired mappings regardless of DM loss configuration. (b) NCE accuracy drops from ∼87% (Phase 1) to ∼50% (Phase 2) as distributions become harder to distinguish after the DM loss activates. (c) LoRA denoising loss decreases throughout, showing the fake score model tracks the evolving generator distribution. (d) Output standard deviation remains above 0.35 for both schedules, confirming no mean collapse occurs. Dashed gray line marks the Phase 1/Phase 2 transition at step 5,000.

C. Qualitative Samples

Figure 6 compares teacher and student outputs. The teacher (100-step DDPM) produces recognizable CIFAR-10 images. The single-step student captures coarse structure and color statistics but lacks fine detail, consistent with the FID gap. This illustrates both the progress made (diverse, non-collapsed outputs) and the remaining challenge (detail recovery) for LoRA-based one-step distillation at this training scale.

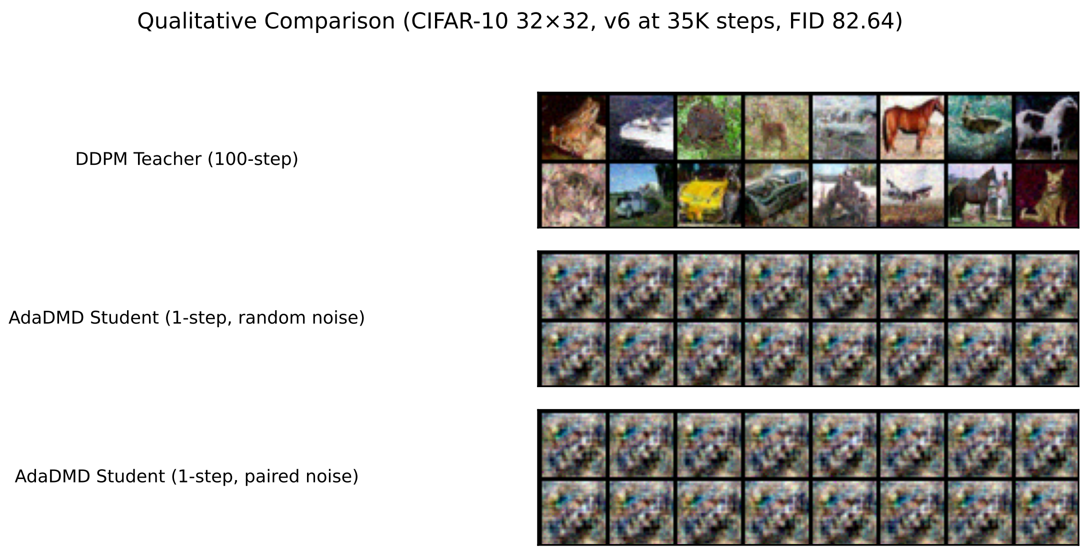

Figure 6. Qualitative comparison on CIFAR-10 32×32 (best model, v6 at 35K steps, FID 82.64). Top: 100-step DDPM teacher outputs. Middle: AdaDMD single-step outputs from random noise. Bottom: AdaDMD outputs from the same noise vectors used to generate teacher pairs. The student captures coarse structure and color statistics while maintaining diversity.